03 - Feature correlation analysis

November 3, 2026

1 Introduction

This document presents the feature correlation analysis, performed on the `FMA_small` dataset [1]. The feature set is presented in the section 3 of the document named "First model comparison". Another document, named "Computational time analysis", presents the computational time needed to compute such a set. This analysis is motivated by an attempt to reduce the computational time of the feature extraction. It is supplemented by an ablation analysis, presented in the document "Ablation analysis".

In this analysis, the correlations are investigated on a feature family level, and then on a individual feature level. The correlations have not been investigated on a feature statistic level, because the computational time of the various statistics for each feature is negligible.

2 Results

In order to evaluate the feature correlation, the Spearman correlation is used. As a form or verification, the Pearson correlation has been used as well. Since the results of Spearman and Person correlations both agree, only the results for the Spearman correlation are presented in this document.

2.1 Feature family correlation

To perform this analysis, the features have been gather into families. The families and the features they gather as described as following:

- **chroma:** `chroma_stft`, `chroma_cqt`, `chroma_cens`, `delta_chroma_stft`, `delta_chroma_cqt`
- **tonnetz:** `tonnetz_cqt`, `tonnetz_cens`
- **mfcc:** `mfcc`, `delta_mfcc`, `delta2_mfcc`
- **spectral_shape:** `spectral_centroid`, `spectral_bandwidth`, `spectral_contrast`, `spectral_flatness`, `spectral_flux`, `spectral_rolloff_50`, `spectral_rolloff_85`, `spectral_rolloff_95`
- **energy_dynamics:** `rms`, `zcr`
- **harmonic_percussive:** `harmonic_rms`, `percussive_rms`, `harmonic_ratio`, `percussive_ratio`
- **rhythm:** `onset_strength`, `tempogram_mean`, `beat_interval`, `tempo`, `beat_rate`

	chroma	tonnetz	mfcc	spectral	en_dyn	HPSS	rhythm
chroma	0.14	0.11	0.12	0.14	0.12	0.16	0.09
tonnetz	0.11	0.18	0.07	0.11	0.08	0.12	0.06
mfcc	0.12	0.07	0.20	0.16	0.16	0.18	0.16
spectral	0.14	0.11	0.16	0.24	0.22	0.24	0.13
en_dyn	0.12	0.08	0.16	0.22	0.29	0.28	0.13
HPSS	0.16	0.12	0.18	0.24	0.28	0.37	0.14
rhythm	0.09	0.06	0.16	0.13	0.13	0.14	0.29

Table 1: This table presents the results of the absolute mean Spearman correlation ($|r|$) between families. The diagonal figures give the average feature correlation within a family. "spectral" is short for the family "spectral_shape", "en_dyn" for "energy_distribution" and "HPSS" for "harmonic_percussive".

Family	Spearman mean $|r|$ within family
Harmonic_percussive	0.37
Energy_dynamics	0.29
rhythm	0.29
spectral_shape	0.24
mfcc	0.20
tonnetz	0.18
chroma	0.14

Table 2: The mean Spearman correlation of features within a family.

The table 1 presents the results of the correlation analysis at feature family level. The absolute Spearman correlation between different families does not exceed 0.28. This means that the families bring different information. So there is no strong evidence to remove a family from the feature set yet. The table 2 gives the mean Spearman correlation of features within a family. The most correlated family is the harmonic_percussive family (HPSS in the table). The correlation between features within this family is 0.37, which suggests that some features in this family are correlated. This will be investigated in the next section.

2.2 Individual feature correlation

After investigating correlation at a family level, the correlation between individual features has been investigated. This level of correlation is the most promising to reduce the computational time. Indeed, the family level is too high: removing a whole family would remove many features and most likely degrade strongly the performances of the classifier. Moreover, as seen in the previous section, the correlations between the different families are not important, so this means that some information would be lost by removing a whole family. This will be checked anyway by performing an ablation analysis. This will be presented in another document. Concerning the feature statistic level, it is too low. Removing some feature statistics from the feature set could be most likely done without impacting too much the classifier performances. But the computational time gain would be very low, since computing feature statistics is very fast.

The figure 1 presents the correlation matrix between individual features. On this figure, there are several light squares, where the correlation is high. The square in the bottom right corresponds to correlation between `beat_interval`, `tempo` and `beat_rate`. Those three are correlated at more than 60%, `tempo` and `beat_rate` being correlated together at 91%. Another correlation square is the one in the center. It corresponds to the spectral_shape family. The correlation is this region varies from 13% to 51%. The last square where the correlation is high is the square center around `harmonic_ratio` and `percussive_ration`, in the harmonic_percussive family. This square extends to the bottom right, in the rhythm family, with the features `onset_strength` and `tempogram_mean`. The highest correlated features in this regions are `harmonic_ratio` and `percussive_ration`, with a correlation of 60%. There are other important cross-family correlations, like `harmonic_ratio` and `percussive_ration` with `rms` and with `spectral_flux` (54% and 46%, and 48% and 55% respectively).

3 Conclusion

This analysis gives some hints on which features could be removed from the feature set. But it does not tell if it is safe to do so without degrading the classifier performances. This can only be checked with an ablation analysis. This will be the next step and be presented in another document.

Thanks to this present analysis, we have some candidates to try in the ablation analysis. The main ones are the harmonic_percussive family. It has an important correlation within itself, but also with other families, like with rhythm, energy_dynamics and spectral_shape. It is the most interesting candidate because this family represents 69% of the computational time. Other candidates to investigate in the ablation procedure are the rhythm family, in particular `beat_interval`, `tempo` and `beat_rate`, the energy_dynamics family, and also shrinking the spectral_shape family.

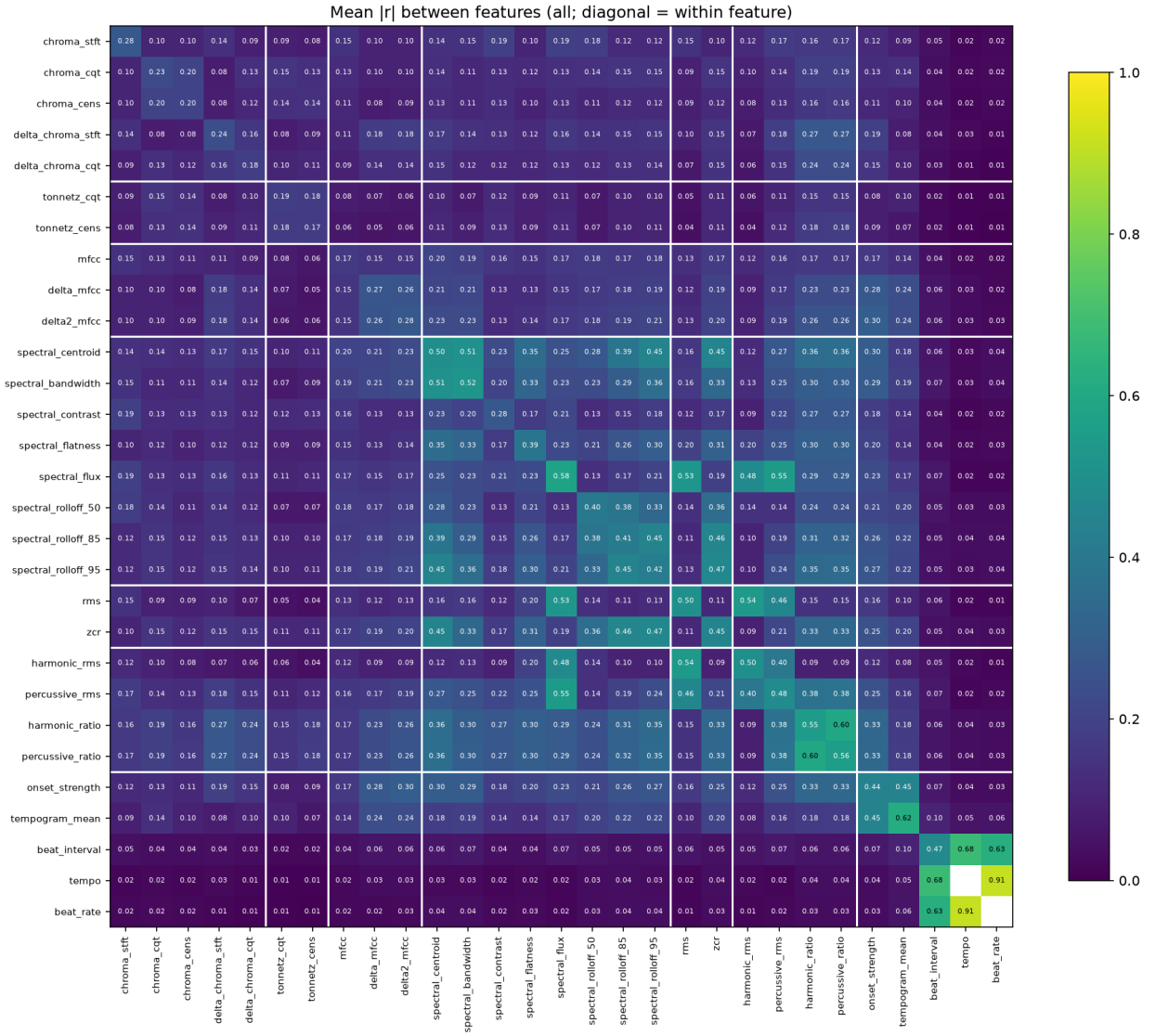


Figure 1: The correlation matrix between individual features. Each cell gives the absoluve Spearman correlation between to two corresponding features. The features are grouped by family. From top to bottom (and from left to right), the families are: chroma, tonnetz, mfcc, spectral_shape, energy_dynamic, harmonic_percussive, rhythm.

References

[1] Michaël Defferrard et al. "FMA: A Dataset for Music Analysis". In: *18th International Society for Music Information Retrieval Conference (ISMIR)*. 2017. arXiv: 1612.01840. URL: https://arxiv.org/abs/1612.01840.